

Team Notebook

[UNI-FIIS] - SaleEnOdeUno

1 Math	2
1.1 ntt	2
1.2 fwht bitwiseconvolution	2
1.3 diophantine	3
1.4 criba	3
1.5 geo2d	3
1.6 xor basis	4
1.7 fft	4
1.8 cht	5
1.9 millerrabin polardrho	5
1.10 primitive root	6
2 Strings	6
2.1 prefix function	6
2.2 suffix array	6
2.3 hashing	6
2.4 z function	7
2.5 manacher	7
3 Graphs	7
3.1 tree bridge	7
3.2 2sat	7
3.3 bellman ford	8
3.4 dinic	8
3.5 lca	8
3.6 maxflow mincost	8
3.7 floyd warshall	9
4 Data Structure	9
4.1 2d fenwick tree	9
4.2 dynamic segment tree	10
4.3 pst	10
4.4 segment tree 2d	10
4.5 merge sort tree	11

4.6 ordered set	11
4.7 sparse table	11
4.8 pst lazy	11
4.9 fenwick tree	12
4.10 centroid decomposition	12
4.11 multi ordered set	12
4.12 hld	12
4.13 segment tree	13
4.14 hld edge	13

1 Math

1.1 ntt

```
const ll MOD = 998244353;
vector<vector<int>> w;
int pow2, root;
int smod(int a, int b, int mod) {return (a + b >= mod ? a + b - mod : a + b);}
int rmod(int a, int b, int mod) {return (a - b < 0 ? a - b + mod : a - b);}
int mult(int a, int b, int mod) {return (ll)a * b % mod;}
int qp(int a, int e, int mod) {
    int ans = 1;
    while (e) {
        if (e & 1) ans = mult(ans, a, mod);
        a = mult(a, a, mod);
        e >>= 1;
    }
    return ans;
}
void calc(int log, int mod) {
    w.resize(log + 1);
    w[0].resize(1, 1);
    for (int l = 1, len = 2; l <= log; l++, len <= 1) {
        w[l].resize(len / 2);
        int wn = qp(root, pow2 / len, mod);
        for (int j = 0; 2 * j < len; j++) {
            if (j & 1) w[l][j] = mult(w[l - 1][j / 2], wn, mod);
            else w[l][j] = w[l - 1][j / 2];
        }
    }
}
void findroot(int mod) {
    pow2 = 1;
    int k = mod - 1;
    while (!(k & 1)) pow2 <= 1, k >>= 1;
    root = 1;
    while (qp(root, pow2, mod) != 1 || qp(root, pow2 / 2, mod) == 1) root++;
}
void ntt(vector<int> &a, bool invert, int mod) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int l = 1, len = 2; len <= n; l++, len <= 1)
        for (int i = 0; i < n; i += len)
            for (int j = 0; 2 * j < len; j++) {
                int u = a[i + j], v = mult(a[i + j + len /
```

```
2], w[l][j], mod);
                a[i + j] = smod(u, v, mod);
                a[i + j + len / 2] = rmod(u, v, mod);
            }
        }
        if (invert) {
            reverse(a.begin() + 1, a.end());
            int invn = qp(n, mod - 2, mod);
            for (int i = 0; i < n; i++) a[i] = mult(a[i], invn, mod);
        }
    }
    vector<int> polymultmod(vector<int> a, vector<int> b, int mod = MOD) {
        int n = 1, log = 0, rn = a.size() + b.size() - 1;
        while (n < a.size() + b.size() - 1) n <= 1, log++;
        findroot(mod);
        calc(log, mod);
        a.resize(n), b.resize(n);
        ntt(a, 0, mod), ntt(b, 0, mod);
        for (int i = 0; i < n; i++) a[i] = mult(a[i], b[i], mod);
        ntt(a, 1, mod);
        a.resize(rn);
        return a;
    }
}
```

1.2 fwht bitwiseconvolution

```
//OR-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <= 1, h <= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) a[j + h] += a[j];
}
void ifwht(vector<ll> &a) {
    int n = A.size();
    for (int s = n, h = s >> 1; s >= 2; s >>= 1, h >>= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) A[j + h] -= A[j];
}
vector<ll> OrConvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
vector<ll> SubsetConvolution(vector<ll> a, vector<ll> b) {
    int n = 1, bit = 1;
```

```
while (n < a.size() || n < b.size()) n <= 1, bit++;
a.resize(n), b.resize(n);
vector<vector<ll>> A(bit + 1, vector<ll>(n)), B(bit + 1, vector<ll>(n));
for (int i = 0; i < n; i++) {
    int f = __builtin_popcount(i);
    A[f][i] = a[i], B[f][i] = b[i];
}
for (int i = 0; i <= bit; i++) fwht(A[i]), fwht(B[i]);
vector<ll> conv(n, 0);
for (int k = 0; k < n; k++) {
    int f = __builtin_popcount(k);
    for (int i = 0; i <= f; i++) conv[k] += A[i][k] * B[f - i][k];
}
ifwht(conv);
return conv;
}
//AND-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <= 1, h <= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) a[j] += a[j + h];
}
void ifwht(vector<ll> &a) {
    int n = A.size();
    for (int s = n, h = n >> 1; s >= 2; s >>= 1, h >>= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) A[j] -= a[j + h];
}
vector<ll> Andconvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
//XOR-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <= 1, h <= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) {
                ll a1 = a[j + h], a0 = a[j];
                a[j + h] = a0 - a1, a[j] = a0 + a1;
            }
}
void ifwht(vector<ll> &a) {
    fwht(A);
```

```

int n = A.size();
for (int i = 0; i < n; i++) A[i] /= n;
}
vector<ll> Xorconvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}

```

1.3 diophantine

```

ll extended_gcd(ll a, ll b, ll &x, ll &y) {
    x = 1, y = 0;
    ll x1 = 0, y1 = 1;
    while (b) {
        ll q = a / b;
        tie(x, x1) = make_pair(x1, x - q * x1);
        tie(y, y1) = make_pair(y1, y - q * y1);
        tie(a, b) = make_pair(b, a - q * b);
    }
    return a < 0 ? (x *= -1, y *= -1, a *= -1, a) : a;
}
bool diophantine(ll a, ll b, ll c, ll &x, ll &y, ll &g) {
    if (a == 0 && b == 0) return c ? 0 : (x = y = 0, 1);
    g = extended_gcd(a, b, x, y);
    return (c % g == 0 ? (x *= c / g, y *= c / g, 1) : 0);
}
ll divdown(ll a, ll b) {
    return a / b - ((a ^ b) < 0 && a % b);
}
ll divup(ll a, ll b) {
    return a / b + ((a ^ b) >= 0 && a % b);
}
vector<pll> allsolution(ll a, ll b, ll c, ll lx, ll rx, ll ly, ll ry) {
    // x = x0 + k*(b/g)    y = y0 - k*(a/g)
    ll x0, y0, g;
    if (!diophantine(a, b, c, x0, y0, g)) return {};
    vector<pll> ans;
    if (!g) {
        for (ll x = lx; x <= rx; x++)
            for (ll y = ly; y <= ry; y++) ans.pb({x, y});
        return ans;
    }
    if (!a) {
        ll y = c / b;
        if (!(ly <= y && y <= ry)) return {};
        for (ll x = lx; x <= rx; x++) ans.pb({x, y});
        return ans;
    }
}

```

```

if (!b) {
    ll x = c / a;
    if (!(lx <= x && x <= rx)) return {};
    for (ll y = ly; y <= ry; y++) ans.pb({x, y});
    return ans;
}
ll dx = b / g, dy = a / g;
ll kmin, kmax;
if (dx > 0) kmin = divup(lx - x0, dx), kmax = divdown(rx - x0, dx);
else kmin = divup(rx - x0, dx), kmax = divdown(lx - x0, dx);
if (dy > 0) kmin = max(kmin, divup(y0 - ry, dy)), kmax = min(kmax, divdown(y0 - ly, dy));
else kmin = max(kmin, divup(y0 - ly, dy)), kmax = min(kmax, divdown(y0 - ry, dy));
for (ll k = kmin; k <= kmax; k++) ans.pb({x0 + k * dx, y0 - k * dy});
return ans;
}
bool positive_sol_minf(ll a, ll b, ll c, ll &x, ll &y, ll cx, ll cy) {
    // f(x,y)= cx*x + cy*y
    ll g;
    if (!diophantine(a, b, c, x, y, g)) return 0;
    ll dx = b / g, dy = a / g;
    ll kmin = -INF, kmax = INF;
    dx > 0 ? kmin = max(kmin, divup(-x, dx)) : kmax = min(kmax, divdown(-x, dx));
    dy < 0 ? kmin = max(kmin, divup(y, dy)) : kmax = min(kmax, divdown(y, dy));
    if (kmin > kmax) return 0;
    ll df = cx * dx - cy * dy;
    ll k = df >= 0 ? kmin : kmax; // df<0 para max
    x += dx * k;
    y -= dy * k;
    return 1;
}
}

```

1.4 criba

```

bool criba[MX];
ll primerfactor[MX], phi[MX];
// criba para primerfactor, phi 0(nlog(log(n)))
void completar() {
    for (int i = 0; i < MX; i++) {
        criba[i] = 1;
        primerfactor[i] = i;
        phi[i] = i;
    }
    criba[0] = 0;
    criba[1] = 0;
    for (ll i = 2; i < MX; i++) {
        if (criba[i]) {
            phi[i]--;

```

```

for (ll j = 2 * i; j < MX; j += i) {
    criba[j] = 0;
    phi[j] -= phi[j] / i;
    if (primerfactor[j] == j) primerfactor[j] = i;
}
}
}
vector<pll> factorizacioncriba(ll n) { // 0(log(n))
    vector<pll> ans;
    while (n > 1) {
        ll f = primerfactor[n];
        ll pot = 0;
        while (primerfactor[n] == f) {
            n /= f;
            pot++;
        }
        ans.pb({f, pot});
    }
    return ans;
}
vector<ll> divisorescriba(ll n) { // 0(d(x))
    vector<ll> fact = {1};
    while (n > 1) {
        ll f = primerfactor[n];
        ll sz = fact.size();
        ll divisor = 1;
        while (n % f == 0) {
            divisor *= f;
            n /= f;
            rep(i, sz) fact.pb(fact[i] * divisor);
        }
        return fact;
    }
}

```

1.5 geo2d

```

struct point {
    ll x, y;
    point() {}
    point(ll _x, ll _y) : x(_x), y(_y) {}
    point operator+(const point &p) const {
        return {x + p.x, y + p.y};
    }
    point operator-(const point &p) const {
        return {x - p.x, y - p.y};
    }
    bool operator==(const point &p) const {
        return x == p.x && y == p.y;
    }
    bool operator<(const point &p) const {
        if (x == p.x) return y < p.y;
        return x < p.x;
    }
}

```

```

    }
};
// Producto punto
long long dot(const point &a, const point &b) {
    return a.x * b.x + a.y * b.y;
}
// Producto cruzado (determinante)
long long cross(const point &a, const point &b) {
    return a.x * b.y - a.y * b.x;
}
// Cross entre vectores AB y AC
long long cross(const point &a, const point &b, const point &c) {
    return (b.x - a.x) * (c.y - a.y) - (b.y - a.y) * (c.x - a.x);
}
// El punto p está en el segmento ab?
bool onSegment(const point &a, const point &b, const point &p) {
    if (cross(a, b, p) != 0) return false;
    return min(a.x, b.x) <= p.x && p.x <= max(a.x, b.x) &&
        min(a.y, b.y) <= p.y &&
        p.y <= max(a.y, b.y);
}
// Interseccion de 2 segmentos
bool intersect(const point &p1, const point &q1, const point &p2, const point &q2) {
    long long d1 = cross(p1, q1, p2);
    long long d2 = cross(p1, q1, q2);
    long long d3 = cross(p2, q2, p1);
    long long d4 = cross(p2, q2, q1);
    if ((d1 > 0 && d2 < 0 || d1 < 0 && d2 > 0) && (d3 > 0 && d4 < 0 || d3 < 0 && d4 > 0))
        return true;
    if (d1 == 0 && onSegment(p1, q1, p2)) return true;
    if (d2 == 0 && onSegment(p1, q1, q2)) return true;
    if (d3 == 0 && onSegment(p2, q2, p1)) return true;
    if (d4 == 0 && onSegment(p2, q2, q1)) return true;
    return false;
}
bool pointInPolygon(const vector<point> &poly, const point &p) {
    bool inside = false;
    int n = poly.size();
    for (int i = 0, j = n - 1; i < n; j = i++) {
        const point &a = poly[i];
        const point &b = poly[j];
        if (onSegment(a, b, p)) return true;
        if ((a.y > p.y) != (b.y > p.y)) {
            long long x_intersection = (b.x - a.x) * (p.y - a.y) - (b.y - a.y) * (p.x - a.x);
            long long y_diff = b.y - a.y;
            bool right = (x_intersection * y_diff) < 0;
            if (right) inside = !inside;
        }
    }
}

```

```

    }
    return inside;
}

```

1.6 xor basis

```

template <typename T, int LOG>
struct Basis {
    T basis[LOG] = {0};
    int dim = 0;
    bool insert(T x) { // Formará parte del Basis?
        for (int i = LOG - 1; i >= 0; i--)
            if ((x >> i) & 1) {
                if (basis[i] x ^= basis[i];
                else {
                    basis[i] = x, dim++;
                    return true;
                }
            }
        return false;
    }
    T size() {
        return ((T)1 << dim);
    }
    bool find(T x) { // ¿x ∈ span{x1,x2...}?
        for (int i = LOG - 1; i >= 0; i--) x = min(x, x ^ basis[i]);
        return x == 0;
    }
    T max_xor(T x = 0) { // max (x^a / a ∈ span{x1,x2...})
        for (int i = LOG - 1; i >= 0; i--) x = max(x, x ^ basis[i]);
        return x;
    }
    T kth(T k) { // The k_smaller of span{x1,x2...}
        if (k < 1 || k > ((T)1 << dim)) return -1;
        T x = 0, cont = ((T)1 << dim);
        for (int i = LOG - 1; i >= 0; i--)
            if (basis[i]) {
                if (k > (cont >> 1)) {
                    if (!(x >> i) & 1) x ^= basis[i];
                    k -= (cont >> 1);
                } else if ((x >> i) & 1) x ^= basis[i];
                cont >>= 1;
            }
        return x;
    }
}

```

1.7 fft

```

typedef complex<double> cd;
vector<vector<cd>> w;
void calc(int log) {
    w.resize(log + 1);
}

```

```

w[0].resize(1, 1);
for (int l = 1, len = 2; l <= log; l++, len <= len * 2) {
    w[l].resize(len / 2);
    cd wn(cos(2 * PI / len), sin(2 * PI / len));
    for (int j = 0; 2 * j < len; j++)
        w[l][j] = (j & 1 ? w[l - 1][j / 2] * wn : w[l - 1][j / 2]);
}
}
void fft(vector<cd> &a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int l = 1, len = 2; len <= n; l++, len <= len * 2)
        for (int i = 0; i < n; i += len)
            for (int j = 0; 2 * j < len; j++) {
                cd u = a[i + j], v = a[i + j + len / 2] *
                    (j & 1 ? 1 : -1);
                a[i + j] = u + v;
                a[i + j + len / 2] = u - v;
            }
    if (invert) {
        reverse(a.begin() + 1, a.end());
        for (int i = 0; i < n; i++) a[i] /= n;
    }
}
vector<ll> polymult(vector<ll> &a, vector<ll> &b) {
    int n = 1, log = 0;
    while (n < a.size() + b.size() - 1) n <= n * 2, log++;
    calc(log);
    vector<cd> A(n);
    for (int i = 0; i < n; i++) A[i] = cd(i < a.size() ? a[i] : 0, i < b.size() ? b[i] : 0);
    fft(A, 0);
    for (int i = 0; i < n; i++) A[i] *= A[i];
    fft(A, 1);
    vector<ll> ans(a.size() + b.size() - 1);
    for (int i = 0; i < ans.size(); i++) ans[i] = round(A[i].imag() / 2);
    return ans;
}
// 2D-FFT
void fft2D(vector<vector<cd>> &a, bool invert) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) fft(a[i], invert);
    vector<cd> b(n);
    for (int j = 0; j < m; j++) {
        for (int i = 0; i < n; i++) b[i] = a[i][j];
        fft(b, invert);
        for (int i = 0; i < n; i++) a[i][j] = b[i];
    }
}

```

```

}
vector<vector<ll>> convolution2D(vector<vector<ll>> &a,
vector<vector<ll>> &b) {
    int n = 1, logn = 0;
    while (n < a.size() + b.size() - 1) n <= 1, logn++;
    int m = 1, logm = 0;
    while (m < a[0].size() + b[0].size() - 1) m <= 1, logm+
+;
    calc(max(logn, logm));
    vector<vector<cd>> A(n, vector<cd>(m));
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            A[i][j] = cd(i < a.size() && j < a[0].size() ?
a[i][j] : 0,
                                i < b.size() && j < b[0].size() ?
b[i][j] : 0);
    fft2D(A, 0);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) A[i][j] *= A[i][j];
    fft2D(A, 1);
    vector<vector<ll>> ans(a.size() + b.size() - 1,
vector<ll>(a[0].size() + b[0].size() - 1));
    for (int i = 0; i < ans.size(); i++)
        for (int j = 0; j < ans[0].size(); j++) ans[i][j] =
round(A[i][j].imag() / 2);
    return ans;
}

```

1.8 cht

```

struct Line {
    mutable ll a, b, c;

    bool operator<(Line r) const {
        return a < r.a;
    }
    bool operator<(ll x) const {
        return c < x;
    }
};
// dynamically insert `a*x + b` lines and query for maximum
// at any x all operations have complexity O(log N)
struct LineContainer : multiset<Line, less<>> {
    ll div(ll a, ll b) {
        return a / b - ((a ^ b) < 0 && a % b);
    }

    bool isect(iterator x, iterator y) {
        if (y == end()) return x->c = INF, 0;
        if (x->a == y->a) x->c = x->b > y->b ? INF : -INF;
        else x->c = div(y->b - x->b, x->a - y->a);
        return x->c >= y->c;
    }

    void add(ll a, ll b) {

```

```

// a *= -1, b *= -1 // for min
auto z = insert({a, b, 0}), y = z++, x = y;
while (isect(y, z)) z = erase(z);
if (x != begin() && isect(--x, y)) isect(x, y =
erase(y));
while ((y = x) != begin() && (--x)->c >= y->c)
isect(x, erase(y));
}

ll query(ll x) {
    if (empty()) return -INF; // INF for min
    auto l = *lower_bound(x);
    return l.a * x + l.b;
    // return -l.a * x - l.b; // for min
}
};

```

1.9 millerrabin polardrho

```

ll mult(ll a, ll b, ll mod) {
    return (int128)a * b % mod;
}
ll qp(ll a, ll e, ll mod) {
    ll ans = 1;
    while (e) {
        if (e & 1) ans = mult(ans, a, mod);
        a = mult(a, a, mod);
        e >>= 1;
    }
    return ans;
}
bool compositetest(ll a, ll d, ll s, ll n) {
    ll x = qp(a, d, n);
    if (x == 1 || x == n - 1) return 0;
    for (ll i = 0; i < s - 1; i++) {
        x = mult(x, x, n);
        if (x == n - 1) return 0;
    }
    return 1;
}
bool esprimo(ll n) {
    if (n < 2) return 0;
    ll d = n - 1, s = 0;
    while ((d & 1) == 0) d >>= 1, s++;
    for (ll a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
37}) {
        if (n == a) return 1;
        if (compositetest(a, d, s, n)) return 0;
    }
    return 1;
}
mt19937_64
rng(chrono::steady_clock::now().time_since_epoch().count());
ll random(ll a, ll b) {
    return uniform_int_distribution<ll>(a, b)(rng);
}

```

```

}
ll f(ll x, ll c, ll mod) {
    ll m = mult(x, x, mod);
    return m + c >= mod ? m + c - mod : m + c;
}
ll getfactor(ll n) {
    if (n % 2 == 0) return 2;
    if (esprimo(n)) return n;
    while (true) {
        ll y = random(1, n - 1);
        ll c = random(1, n - 1);
        ll m = 64, g = 1, r = 1, q = 1;
        ll x, ys;
        while (g == 1) {
            x = y;
            for (ll i = 0; i < r; i++) y = f(y, c, n);
            ll k = 0;
            while (k < r && g == 1) {
                ys = y;
                ll lim = min(m, r - k);
                for (ll i = 0; i < lim; i++) {
                    y = f(y, c, n);
                    ll diff = x > y ? x - y : y - x;
                    q = mult(q, diff, n);
                }
                g = __gcd(q, n);
                k += m;
            }
            r <= 1;
        }
        if (g == n) {
            do {
                ys = f(ys, c, n);
                ll diff = x > ys ? x - ys : ys - x;
                g = __gcd(diff, n);
            } while (g == 1);

            if (g > 1 && g < n) return g;
        }
    }
}
vector<pll> fp(ll n) {
    stack<ll> st;
    st.push(n);
    vector<pll> ans;
    map<ll, ll> cont;
    while (!st.empty()) {
        ll x = st.top();
        st.pop();
        if (x == 1) continue;
        if (esprimo(x)) cont[x]++;
        else {
            ll f = getfactor(x);
            st.push(f), st.push(x / f);
        }
    }
}

```

```

}
for (auto [x, f] : cont) ans.pb({x, f});
return ans;
}

```

1.10 primitive root

```

int primitive_root(int p) {
    int n = p - 1; // n=phi(p) -> n=p-1 si p es primo
    vector<int> fp;
    for (int i = 2; i * i <= n; i++)
        if (n % i == 0) {
            fp.pb(i);
            while (n % i == 0) n /= i;
        }
    if (n > 1) fp.pb(n);
    n = p - 1;
    bool ok = 0;
    for (int g = 2; g <= n; g++) {
        bool ok = 1;
        for (int x : fp)
            if (qp(g, n / x, p) == 1) {
                ok = 0;
                break;
            }
        if (ok) return g;
    }
    return 0;
}

```

2 Strings

2.1 prefix function

```

void prefix_function(string &s, int pf[]) {
    int n = s.size(); pf[0] = 0;
    for (int i = 1; i < n; i++) {
        int j = pf[i - 1];
        while (j > 0 && s[i] != s[j]) j = pf[j - 1];
        pf[i] = j + (s[i] == s[j]);
    }
}

```

2.2 suffix array

```

enum Comparison { LESS, EQUAL, GREATER };

Comparison getComparison(int a, int b) {
    if (a < b) return Comparison::LESS;
    if (a > b) return Comparison::GREATER;
    return Comparison::EQUAL;
}

const int ALPH = 256;

```

```

struct SuffixArray {
    vector<int> suffixArray;
    vector<int> lcp;
    // lcp[i] = largest common preffix of sa[i] and sa[i + 1] (in sorted list)
    vector<int> pos;

    vector<int> sortCyclic(string &s) { // O(n log n)
        int n = s.size();
        vector<int> p(n), c(n), cnt(max(ALPH, n), 0);
        // p[] : sorted array with the starting index of substrings of length 2^k
        // c[] : equivalence class
        // counting sort
        for (int i = 0; i < n; i++) cnt[s[i]]++;
        for (int i = 1; i < ALPH; i++) cnt[i] += cnt[i - 1];
        for (int i = 0; i < n; i++) p[--cnt[s[i]]] = i;

        c[p[0]] = 0;
        int classes = 1;
        for (int i = 1; i < n; i++) {
            if (s[p[i]] != s[p[i - 1]]) classes++;
            c[p[i]] = classes - 1;
        }
        // radix sort
        vector<int> pNew(n), cNew(n);
        for (int k = 0; (1 << k) < n; k++) {
            // sort pNew by the second substring
            for (int i = 0; i < n; i++) {
                pNew[i] = p[i] - (1 << k);
                if (pNew[i] < 0) pNew[i] += n;
                cnt[i] = 0;
            }
            // counting sort in the first substring
            for (int i = 0; i < n; i++) cnt[c[pNew[i]]]++;
            for (int i = 1; i < classes; i++) cnt[i] += cnt[i - 1];
            for (int i = n - 1; i >= 0; i--) {
                p[--cnt[c[pNew[i]]]] = pNew[i];
            }
            cNew[p[0]] = 0;
            classes = 1;
            for (int i = 1; i < n; i++) {
                int x = p[i] + (1 << k);
                if (x >= n) x -= n;
                int y = p[i - 1] + (1 << k);
                if (y >= n) y -= n;
                pair<int, int> cur = {c[p[i]], c[x]};
                pair<int, int> prev = {c[p[i - 1]], c[y]};
                if (cur != prev) classes++;
                cNew[p[i]] = classes - 1;
            }
            c = cNew;
        }
        return p;
    }
}

```

```

}

void buildLCP(string &s) { // O(n)
    int n = s.size();
    lcp = vector<int>(n - 1, 0);
    pos = vector<int>(n);
    for (int i = 0; i < n; i++) pos[suffixArray[i]] = i;
    int k = 0;
    for (int r = 0; r < n; r++) {
        if (pos[r] == 0) continue;
        int l = suffixArray[pos[r] - 1];
        while (s[l + k] == s[r + k]) k++;
        lcp[pos[r] - 1] = k;
        k = max(0, k - 1);
    }
}

// Suffix array: contain starting indexes of the all the suffixes (sorted)
SuffixArray(string &s) { // O(n log n)
    char minChar = 'a' - 1;
    s += minChar;
    suffixArray = sortCyclic(s);
    suffixArray.erase(suffixArray.begin());
    s.pop_back();
    buildLCP(s);
}

// return the lcp of the suffixes s[l...n - 1] and s[r...n - 1]
/*int getLCP(int l, int r) {
    assert(l != r);
    int minPos = min(pos[l], pos[r]);
    int maxPos = max(pos[l], pos[r]);
    return min(lcp[minPos], ..., lcp[maxPos - 1])
}*/

/*Comparison compare(int l1, int r1, int l2, int r2) {
    // compare two substrings [l1, r1], [l2, r2].
    int curlCP = (l1 == l2) ? suffixArray.size() : getLCP(l1, l2);
    if (r1 < l1 + curlCP) {
        if (r2 < l2 + curlCP) {
            int sz1 = r1 - l1 + 1;
            int sz2 = r2 - l2 + 1;
            return getComparison(sz1, sz2);
        } else return Comparison::LESS;
    } else if (r2 < l2 + curlCP) return Comparison::GREATER;
    return getComparison(pos[l1], pos[l2]);
}*/
};

```

2.3 hashing

```

struct Hashing{
    const int mod = 1e9+9;
    const int p = 367, q = 467;
    int ep[N],eq[N],prp[N],prq[N];

    int P(int a,int b){ return 1ll*a*b%mod;}
    int S(int a,int b){ return (a+b)%mod;}

    void build(string &s){
        int n = s.size();
        ep[0] = eq[0] = 1;
        prp[0] = prq[0] = 0;
        for (int i=1; i<=n; i++){
            prp[i] = S(P(prp[i-1], p), s[i-1]);
            prq[i] = S(P(prq[i-1], q), s[i-1]);
            ep[i] = P(ep[i-1], p);
            eq[i] = P(eq[i-1], q);
        }
    }

    ll substr(int i,int len){ i++;
        ll hp = S(prp[i+len-1], mod-P(prp[i-1], ep[len]));
        ll hq = S(prq[i+len-1], mod-P(prq[i-1], eq[len]));
        return hp*mod + hq;
    }
} H1, H2;

```

2.4 z function

```

void z_function(string &s,int zf[]){
    int n = s.size();
    for (int i=1,l=0,r=0; i<n; i++){
        if (i < r) zf[i] = min(zf[i-l], r-i);
        while (i+zf[i] < n && s[i+zf[i]] == s[zf[i]]) zf[i]++;
        if (i+zf[i] > r) l = i, r = i+zf[i];
    }
}

```

2.5 manacher

```

// d1[i] odd number of palindromes centered at i
// d2[i] even number of palindromes centered at (i, i + 1)
void manacher(string &s, vector<int> &d1, vector<int> &d2) {
    int n = s.size();
    d1 = vector<int>(n);
    d2 = vector<int>(n, 0);
    int l = -1, r = -1;
    for (int i = 0; i < n; i++) {
        d1[i] = (i > r) ? 1 : min(d1[l + r - i], r - i + 1);
        while (i - d1[i] >= 0 && i + d1[i] < n && s[i - d1[i]] == s[i + d1[i]]) {
            d1[i]++;
        }
        if (i + d1[i] - 1 > r) r = i + d1[i] - 1, l = i - d1[i];
    }
}

```

```

(d1[i] - 1);
}
l = -1, r = -1;
for (int i = 0; i < n - 1; i++) {
    d2[i] = (i >= r) ? 0 : min(d2[l + r - i - 1], r - i);
    while (i - d2[i] >= 0 && i + d2[i] + 1 < n && s[i - d2[i]] == s[i + d2[i] + 1]) {
        d2[i]++;
    }
    if (i + d2[i] > r && d2[i] > 0) r = i + d2[i], l = i - d2[i] + 1;
}
}

```

3 Graphs

3.1 tree bridge

```

vector<int> adj[MX];
int tin[MX], low[MX], timer;
vector<pii> bridges;
int component[MX];
vector<int> nods[MX];
vector<array<int, 3>> adjcomp[MX];
void dfs(int u, int p) {
    tin[u] = low[u] = ++timer;
    for (int v : adj[u])
        if (v != p) {
            if (tin[v]) low[u] = min(low[u], tin[v]);
            else dfs(v, u), low[u] = min(low[u], low[v]);
            if (tin[u] < low[v]) bridges.pb({u, v});
        }
}
bool isbridge(int u, int v) {
    if (tin[u] > tin[v]) swap(u, v);
    return tin[u] < low[v];
}
void dfsc(int u, int comp) {
    component[u] = comp;
    nods[comp].pb(u);
    for (int v : adj[u])
        if (!isbridge(u, v) && !component[v]) dfsc(v, comp);
}
int tree_bridge(int n) {
    dfs(1, 1);
    int comp = 0;
    for (int u = 1; u <= n; u++)
        if (!component[u]) dfsc(u, ++comp);
    for (auto [u, v] : bridges) {
        int cu = component[u], cv = component[v];
        adjcomp[cu].pb({cv, u, v});
        adjcomp[cv].pb({cu, u, v});
    }
}

```

```

return comp;
}

```

3.2 2sat

```

struct SAT { // Index_1
    vector<vector<int>> adj, adjt;
    vector<bool> used;
    vector<int> comp;
    int n;
    stack<int> st;
    SAT(int n) : n(n) {
        adj.resize(2 * n + 1), adjt.resize(2 * n + 1);
        used.assign(2 * n + 1, 0), comp.assign(2 * n + 1, 0);
    }
    void add(int p, bool bp, int q, bool bq) { // add p -> q
        p += (!bp) * n;
        q += (!bq) * n;
        adj[p].pb(q);
        adjt[q].pb(p);
    }
    void addor(int p, bool bp, int q, bool bq) { // add p or q
        add(p, !bp, q, bq);
        add(q, !bq, p, bp);
    }
    void addxor(int p, bool bp, int q, bool bq) { // add p xor q
        addor(p, bp, q, bq);
        addor(p, !bp, q, !bq);
    }
    void addnxor(int p, bool bp, int q, bool bq) { // add ~(p xor q)
        addor(p, !bp, q, bq);
        addor(p, bp, q, !bq);
    }
    void dfs(int u) {
        used[u] = 1;
        for (int v : adj[u])
            if (!used[v]) dfs(v);
        st.push(u);
    }
    void _dfs(int u, int c) {
        comp[u] = c;
        for (int v : adjt[u])
            if (!comp[v]) _dfs(v, c);
    }
    bool solve(vector<bool> &ans) {
        ans.resize(n + 1);
        for (int u = 1; u <= 2 * n; u++)
            if (!used[u]) dfs(u);
        int c = 0;
        while (!st.empty()) {
            int u = st.top();

```



```

    st.pop();
    if (!comp[u]) _dfs(u, ++c);
}
for (int p = 1; p <= n; p++) {
    if (comp[p] == comp[p + n]) return false;
    ans[p] = comp[p] > comp[p + n];
}
return true;
}
};

```

3.3 bellman ford

```

// find distance from source node to all nodes.
// supports negative edge weights.
// returns true if a negative cycle is detected.
bool bellmanFord(int s, int n) { // O(VE)
    d.assign(n + 1, INF);
    vector<bool> cycle(n, false);
    bool f = false;
    for (int i = 0; i < n; i++) {
        f = false;
        for (int u = 0; u < n; u++) {
            for (auto [v, w] : adj[u]) {
                if (d[u] != INF && d[u] + w < d[v]) {
                    d[v] = d[u] + w;
                    parent[v] = u;
                    f = true;
                    if (i == n - 1) cycle[v] = true;
                }
            }
        }
        if (!f) return false;
    }
    return true;
}

```

3.4 dinic

```

// This implementation is for small graphs G(V, E) (|V| <= 3000 aprox)
// where O(|V|^2) memory will fit the memory limit.
using Cap = long long;
const int MX = 3000;
const Cap INF = 1e18;
struct Graph {
    vector<int> adj[MX];
    Cap cap[MX][MX];
    Cap flow[MX][MX];
    bool added[MX][MX];
    int level[MX];
    int nextEdge[MX];

    void clear(int n) {
        for (int i = 0; i < n; i++) {

```

```

            adj[i].clear();
            for (int j = 0; j < n; j++) cap[i][j] = 0,
            flow[i][j] = 0, added[i][j] = false;
        }
    }

    void addEdge(int u, int v, Cap w, bool dir) {
        if (!added[u][v]) {
            adj[u].push_back(v);
            adj[v].push_back(u);
            added[u][v] = added[v][u] = true;
        }
        cap[u][v] += w;
        if (!dir) cap[v][u] += w;
    }

    Cap dfs(int u, int t, Cap f) { // O(E)
        if (u == t) return f;
        for (int &i = nextEdge[u]; i < adj[u].size(); i++) {
            int v = adj[u][i];
            Cap cf = cap[u][v] - flow[u][v];
            if (cf == 0 || level[v] != level[u] + 1)
                continue;

            Cap ret = dfs(v, t, min(f, cf));
            if (ret > 0) {
                flow[u][v] += ret;
                flow[v][u] -= ret;
                return ret;
            }
        }
        return 0;
    }

    bool bfs(int s, int t, int n) { // O(E)
        fill(level, level + n, -1);
        queue<int> q({s});
        level[s] = 0;
        while (!q.empty()) {
            int u = q.front();
            nextEdge[u] = 0;
            q.pop();
            for (int v : adj[u]) {
                Cap cf = cap[u][v] - flow[u][v];
                if (level[v] == -1 && cf > 0) {
                    level[v] = level[u] + 1;
                    q.push(v);
                }
            }
        }
        return level[t] != -1;
    }

    Cap maxFlow(int s, int t, int n) {
        // General: O(E * V^2), O(E * V + V * |F|)
        // Unit Cap: O(E * min(E^(1/2), V^(2/3)))

```

```

        // Unit Network: O(E * V^(1/2))
        // In unit network, all the edges have unit capacity
        // and for any vertex except s and t either the
        // incoming or outgoing edge is unique.
        Cap ans = 0;
        while (bfs(s, t, n)) { // O(V) iterations
            // after bfs, the graph is a DAG
            while (Cap inc = dfs(s, t, INF)) ans += inc;
        }
        return ans;
    }
} G;

```

3.5 lca

```

int tIn[MX], tOut[MX];
int anc[MX][LG];
int timer;

void dfs(int u) {
    tIn[u] = timer++;
    for (int j = 1; j < LG; j++) anc[u][j] = anc[anc[u][j - 1]][j - 1];
    for (int v : adj[u]) {
        if (v == anc[u][0]) continue;
        anc[v][0] = u;
        dfs(v);
    }
    tOut[u] = timer++;
}

bool isAnc(int u, int v) { // is u ancestor of v ?
    return tIn[u] <= tIn[v] && tOut[u] >= tOut[v];
}

int lca(int u, int v) { // O(log n)
    if (isAnc(u, v)) return u;
    if (isAnc(v, u)) return v;
    for (int i = LG - 1; i >= 0; i--) {
        if (!isAnc(anc[u][i], v)) u = anc[u][i];
    }
    return anc[u][0];
}

```

3.6 maxflow mincost

```

using Cap = long long;
using Cost = long long;

const int MX = 5000;
const Cap INF_CAP = 1e18;
const Cost INF_COST = 1e18;

struct Edge {
    int to;

```



```

Cap flow, cap;
Cost cost;
int rev; // index of the backward edge in the adj list
of to
Edge(int to, Cap cap, Cost cost, int rev)
: to(to), flow(0), cap(cap), cost(cost), rev(rev) {}
};

struct Graph {
vector<Edge> adj[MX];
int parentEdge[MX];
Cost pot[MX];
//|pot[u]| <= the maximum sum of absolute cost in a path
// which is also bounded by [(V - 1) * C]
// where C is the maximum value of |cost(e)| for all
edges e
void clear(int n) {
for (int i = 0; i < n; i++) {
adj[i].clear();
pot[i] = 0;
}
}

void addEdge(int u, int v, Cap w, Cost cost, bool dir) {
adj[u].push_back(Edge(v, w, cost, adj[v].size()));
adj[v].push_back(Edge(u, 0, -cost, adj[u].size() -
1));
if (!dir) addEdge(v, u, w, cost, true);
}

void spfa(int s, int n) { // O(E V)
vector<bool> inQueue(n, false);
for (int i = 0; i < n; i++) pot[i] = INF_COST;
queue<int> q;
pot[s] = 0;
inQueue[s] = true;
q.push(s);
while (!q.empty()) {
int u = q.front();
q.pop();
inQueue[u] = false;
for (Edge e : adj[u]) {
int v = e.to;
if (e.cap - e.flow > 0 && pot[u] + e.cost <
pot[v]) {
pot[v] = pot[u] + e.cost;
if (!inQueue[v]) q.push(v);
inQueue[v] = true;
}
}
}

void pushFlow(int s, int t, Cap inc) {
int v = t;

```

```

while (v != s) {
Edge &backward = adj[v][parentEdge[v]];
int u = backward.to;
Edge &forward = adj[u][backward.rev];
forward.flow += inc;
backward.flow -= inc;
v = u;
}
}

pair<Cap, Cost> dijkstra(int s, int t, int n) { // O(E
log V)
//<flow, cost>
using Path = pair<Cost, int>; //<weight, node>
priority_queue<Path, vector<Path>, greater<Path>> q;
vector<Cost> d(n, INF_COST);
vector<Cap> residualCap(n, 0);
d[s] = 0;
residualCap[s] = INF_CAP;
q.push(Path(d[s], s));
while (!q.empty()) {
auto [dist, u] = q.top();
q.pop();
if (dist != d[u]) continue;
for (Edge e : adj[u]) {
int v = e.to;
Cap cf = e.cap - e.flow;
Cost cost = e.cost + pot[u] - pot[v];
if (cf > 0 && d[u] + cost < d[v]) {
d[v] = d[u] + cost;
q.push(Path(d[v], v));
residualCap[v] = min(residualCap[v],
cf);
parentEdge[v] = e.rev;
}
}
}
if (d[t] == INF_COST) return {0, 0};
for (int i = 0; i < n; i++) {
if (pot[i] != INF_COST) pot[i] += d[i];
}
Cap cf = residualCap[t];
pushFlow(s, t, cf);
return {cf, pot[t] * cf};
}

pair<Cap, Cost> minCostFlow(int s, int t, int n) {
// O(E log V * maxFlow)
// maxFlow <= V * U, where U is the maximum capacity
// Assumption: Initially no negative cycles
//<maxFlow, minCost>
spfa(s, n); // not necessary if there is no negative
edges
pair<Cap, Cost> inc;
Cap flow = 0;

```

```

Cost cost = 0;
do {
inc = dijkstra(s, t, n);
flow += inc.first;
cost += inc.second;
} while (inc.first > 0);
return {flow, cost};
}
} G;

```

3.7 floyd warshall

```

void Ini(int n) {
for0(u, n) {
for0(v, n) d[u][v] = INF, parent[u][v] = -1;
d[u][u] = 0;
}
}

void addEdge(int u, int v, int w) {
if (w < d[u][v]) {
d[u][v] = w;
parent[u][v] = u;
}
}

void floydWarshall(int n) { // O(V^3)
for0(k, n) for0(u, n) for0(v, n) {
if (d[u][k] == INF) continue;
if (d[k][v] == INF) continue;
if (d[u][k] + d[k][v] < d[u][v]) {
d[u][v] = max(-INF, d[u][k] + d[k][v]);
parent[u][v] = parent[k][v];
}
}

// negative cycles
for0(u, n) for0(v, n) for0(k, n) {
if (d[k][k] < 0 && d[u][k] != INF && d[k][v] != INF)
{
d[u][v] = -INF;
}
}
}

```

4 Data Structure

4.1 2d fenwick tree

```

struct BIT2D {
vector<vector<int>> t;
int n, m;
BIT2D(int n, int m) : n(n), m(m) {
t.assign(n, vector<int>(m, 0));
}
}

```

```

}
void update(int x, int y, int add) {
    for (int i = x; i < n; i |= i + 1)
        for (int j = y; j < m; j |= j + 1) t[i][j] +=
add;
}
int sum(int x, int y) {
    int ans = 0;
    for (int i = x; i >= 0; i = (i & (i + 1)) - 1)
        for (int j = y; j >= 0; j = (j & (j + 1)) - 1)
ans += t[i][j];
    return ans;
}
int query(int lx, int rx, int ly, int ry) {
    return sum(rx, ry) - sum(lx - 1, ry) - sum(rx, ly -
1) + sum(lx - 1, ly - 1);
}
};

```

4.2 dynamic segment tree

```

struct nod {
    ll t;
    ll lazy;
    nod *l;
    nod *r;
};
static nod n[ND];
static int id = 0;
nod *crearnodo() {
    n[id].t = NEUT, n[id].lazy = NEUT_LAZY, n[id].l
= n[id].r = NULL;
    return &n[id++];
}
ll f(ll a, ll b) {
    return a + b;
}
void aply(ll val, nod *u, ll l, ll r) {
    u->t = (r - l + 1) * val;
    u->lazy = val;
}
struct segmentTree {
    ll n;
    nod *root = NULL;
    segmentTree(ll n) : n(n) {}
    void push(nod *u, ll l, ll r) {
        if (u->lazy == NEUT_LAZY || l == r) return;
        ll m = (l + r) / 2;
        if (!u->l) u->l = creaknodo();
        if (!u->r) u->r = creaknodo();
        aply(u->lazy, u->l, l, m);
        aply(u->lazy, u->r, m + 1, r);
        u->lazy = NEUT_LAZY;
    }
    ll query(ll ql, ll qr, nod *u, ll l, ll r) {

```

```

        if (qr < l || r < ql || !u) return NEUT;
        if (ql <= l && r <= qr) return u->t;
        push(u, l, r);
        ll m = (l + r) / 2;
        return f(query(ql, qr, u->l, l, m), query(ql, qr, u-
>r, m + 1, r));
    }
    ll query(ll ql, ll qr) {
        return query(ql, qr, root, 0, n - 1);
    }
    void update(ll ql, ll qr, ll val, nod *u, ll l, ll r) {
        if (qr < l || r < ql) return;
        if (!u) u = creaknodo();
        if (ql <= l && r <= qr) aply(val, u, l, r);
        else {
            push(u, l, r);
            ll m = (l + r) / 2;
            update(ql, qr, val, u->l, l, m);
            update(ql, qr, val, u->r, m + 1, r);
            u->t = f(u->l ? u->l->t : NEUT, u->r ? u->r->t :
NEUT);
        }
    }
    void update(ll ql, ll qr, ll val) {
        update(ql, qr, val, root, 0, n - 1);
    }
};

```

4.3 pst

```

const ll MX = 2e5 + 5;
const ll Q = 2e5 + 5;
const ll MXT = 2 * MX + 20 * Q;
struct PST {
    ll root[Q + 1], t[MXT], L[MXT], R[MXT];
    ll n, nodos, lastversion = 0;
    PST(vector<ll> &a) : n(a.size()) {
        nodos = 2 * n - 1;
        root[0] = 1;
        build(a, 1, 0, n - 1);
    }
    ll f(ll a, ll b) {
        return a + b;
    }
    void build(vector<ll> &a, ll id, ll l, ll r) {
        if (l == r) t[id] = a[l];
        else {
            ll m = (l + r) / 2;
            ll idl = id + 1;
            ll idr = id + 2 * (m - l + 1);
            L[id] = idl;
            R[id] = idr;
            build(a, idl, l, m);
            build(a, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);

```

```

        }
    }
    ll query(ll ql, ll qr, ll id, ll l, ll r) {
        if (ql <= l && r <= qr) return t[id];
        ll m = (l + r) / 2;
        if (qr <= m) return query(ql, qr, L[id], l, m);
        if (m + 1 <= ql) return query(ql, qr, R[id], m + 1,
r);
        return f(query(ql, qr, L[id], l, m), query(ql, qr,
R[id], m + 1, r));
    }
    ll query(ll ql, ll qr, ll version = -1) {
        if (version == -1) version = lastversion;
        return query(ql, qr, root[version], 0, n - 1);
    }
    void update(ll pos, ll val, ll prev, ll &id, ll l, ll r)
{
        id = ++nodos;
        if (l == r) t[id] = val;
        else {
            ll m = (l + r) / 2;
            L[id] = L[prev];
            R[id] = R[prev];
            if (pos <= m) update(pos, val, L[prev], L[id],
l, m);
            else update(pos, val, R[prev], R[id], m + 1, r);
            t[id] = f(t[L[id]], t[R[id]]);
        }
    }
    void update(ll pos, ll val, ll version = -1) {
        if (version == -1) version = lastversion;
        lastversion++;
        update(pos, val, root[version], root[lastversion],
0, n - 1);
    }
};

```

4.4 segment tree 2d

```

struct ST2D {
    int n, m;
    vector<vector<ll>> t;
    ll f(ll a, ll b) {
        return a + b;
    }
    ST2D(vector<vector<ll>> &a) {
        n = a.size();
        m = a[0].size();
        t.assign(2 * n, vector<ll>(2 * m, 0));
        for (int i = 0; i < n; i++)
            for (int j = 0; j < m; j++) t[i + n][j + m] =
a[i][j];
        for (int i = n; i < 2 * n; i++)
            for (int j = m - 1; j > 0; j--) t[i][j] = f(t[i
][j << 1], t[i][j << 1 | 1]);

```

```

    for (int i = n - 1; i > 0; i--)
        for (int j = 1; j < 2 * m; j++) t[i][j] = f(t[i
<< 1][j], t[i << 1 | 1][j]);
    }
    void update(int x, int y, ll val) {
        x += n, y += m, t[x][y] = val;
        for (int j = y; j > 1; j >= 1) t[x][j >> 1] =
f(t[x][j], t[x][j ^ 1]);
        for (int i = x >> 1; i > 0; i >= 1) {
            int j = y;
            t[i][j] = f(t[i << 1][j], t[i << 1 | 1][j]);
            for (; j > 1; j >= 1) t[i][j >> 1] = f(t[i][j],
t[i][j ^ 1]);
        }
    }
    ll queryY(int i, int l, int r) {
        ll ans = 0;
        for (l += m, r += m + 1; l < r; l >= 1, r >= 1) {
            if (l & 1) ans = f(ans, t[i][l++]);
            if (r & 1) ans = f(t[i][--r], ans);
        }
        return ans;
    }
    ll query(int lx, int rx, int ly, int ry) {
        ll ans = 0;
        for (x1 += n, x2 += n + 1; x1 < x2; x1 >= 1, x2 >=
1) {
            if (x1 & 1) ans = f(ans, queryY(x1++, y1, y2));
            if (x2 & 1) ans = f(queryY(--x2, y1, y2), ans);
        }
        return ans;
    }
};

```

4.5 merge sort tree

```

struct segmentTree {
    vector<vector<ll>> seg;
    ll n, N;
    segmentTree() {}
    segmentTree(vector<ll> &a) : n(a.size()) {
        N = 1;
        while (N < n) N <= 1;
        seg.assign(2 * N, vector<ll>());
        for (int i = 0; i < n; i++) seg[i + N] = {a[i]};
        for (int i = N - 1; i > 0; i--) {
            auto &l = seg[i << 1], &r = seg[i << 1 | 1];
            seg[i].resize((ll)l.size() + (ll)r.size());
            merge(all(l), all(r), seg[i].begin());
        }
    }
    // cantidad de elementos mayores a "a" en el intervalo
[ql,qr]
    ll query1(ll ql, ll qr, ll a) {
        ll l = ql + N, r = qr + N;

```

```

        ll ans = 0;
        while (l <= r) {
            if (l & 1) {
                auto &v = seg[l++];
                ans += v.end() - upper_bound(all(v), a);
            }
            if (!(r & 1)) {
                auto &v = seg[r--];
                ans += v.end() - upper_bound(all(v), a);
            }
            l >= 1;
            r >= 1;
        }
        return ans;
    }
    // cantidad de elementos menores a "a" en el intervalo
[ql,qr]
    ll query2(ll ql, ll qr, ll a) {
        ll l = ql + N, r = qr + N;
        ll ans = 0;
        while (l <= r) {
            if (l & 1) {
                auto &v = seg[l++];
                ans += lower_bound(all(v), a) - v.begin();
            }
            if (!(r & 1)) {
                auto &v = seg[r--];
                ans += lower_bound(all(v), a) - v.begin();
            }
            l >= 1;
            r >= 1;
        }
        return ans;
    }
};

```

4.6 ordered set

```

// find_by_order(k) -> return iterator to the k-th element
(0-indexed) - O(log n)
// order_of_key(num) -> # of items strictly smaller than num
- O(log n)
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;

template <typename T>
using ordered_set =
    tree<T, null_type, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;

ordered_set<int> S;

```

4.7 sparse table

```

const int LG = __lg(MX) + 1;
int st[MX][LG];
void build(vector<int> &A) { // O(|f| n log n)
    int n = A.size();
    for (int i = 0; i < n; i++) st[i][0] = A[i];
    for (int k = 1; k < LG; k++) {
        for (int i = 0; i + (1 << k) <= n; i++) {
            st[i][k] = min(st[i][k - 1], st[i + (1 << (k -
1))][k - 1]);
        }
    }
}
int query(int l, int r) { // O(|f|)
    int lg = __lg(r - l + 1);
    return min(st[l][lg], st[r - (1 << lg) + 1][lg]);
}

```

4.8 pst lazy

```

const int MX = 2e5 + 5;
const int Q = 2e5 + 5;
const int LOG = 20;
const int ND = 2 * MX + 6 * Q * LOG;
const ll NEUT = 0;
const ll NEUT_LAZY = -2 * INF;
struct nod {
    ll t, lazy;
    nod *l, *r;
};
nod n[ND];
int id = 0;
nod *crearnodo(nod *u = NULL) {
    if (u) n[id].t = u->t, n[id].lazy = u->lazy,
n[id].l = u->l, n[id].r = u->r;
    else n[id].t = NEUT, n[id].lazy = NEUT_LAZY,
n[id].l = n[id].r = NULL;
    return &n[id++];
}
ll f(ll a, ll b) {
    return a + b;
}
void apply(ll val, nod *u, ll l, ll r) {
    u->t += (r - l + 1) * val;
    if (u->lazy == NEUT_LAZY) u->lazy = val;
    else u->lazy += val;
}
struct PST {
    nod *root[MX + 5];
    ll n;
    ll lastversion = -1;
    PST(vector<ll> &a) : n(a.size()) {
        root[++lastversion] = crearnodo();
        build(a, root[0], 0, n - 1);
    }
    void build(vector<ll> &a, nod *u, ll l, ll r) {

```

```

    if (!u) u = crearnodo();
    if (l == r) u->t = a[l];
    else {
        ll m = (l + r) >> 1;
        build(a, u->l, l, m);
        build(a, u->r, m + 1, r);
        u->t = f(u->l->t, u->r->t);
    }
}

void push(nod *u, ll l, ll r) {
    if (u->lazy == NEUT_LAZY || l == r) return;
    ll m = (l + r) >> 1;
    u->l = crearnodo(u->l), u->r = crearnodo(u->r);
    apply(u->lazy, u->l, l, m);
    apply(u->lazy, u->r, m + 1, r);
    u->lazy = NEUT_LAZY;
}

nod *update(ll ql, ll qr, ll val, nod *prev, ll l, ll r)
{
    if (qr < l || r < ql) return prev;
    nod *u = crearnodo(prev);
    if (ql <= l && r <= qr) apply(val, u, l, r);
    else {
        push(u, l, r);
        ll m = (l + r) >> 1;
        u->l = update(ql, qr, val, u->l, l, m);
        u->r = update(ql, qr, val, u->r, m + 1, r);
        u->t = f(u->l->t, u->r->t);
    }
    return u;
}

void update(ll ql, ll qr, ll val, ll version = -1) {
    if (version == -1) version = lastversion;
    root[++lastversion] = update(ql, qr, val,
    root[version], 0, n - 1);
}

ll query(ll ql, ll qr, nod *u, ll l, ll r) {
    if (!u || qr < l || r < ql) return NEUT;
    if (ql <= l && r <= qr) return u->t;
    push(u, l, r);
    ll m = (l + r) >> 1;
    return f(query(ql, qr, u->l, l, m), query(ql, qr, u->r, m + 1, r));
}

ll query(ll ql, ll qr, ll version = -1) {
    if (version == -1) version = lastversion;
    return query(ql, qr, root[version], 0, n - 1);
}
};

```

4.9 fenwick tree

```

struct FenwickTree { // Index 1
    int n, ft[MX + 1];
    void update(int l, int v) {

```

```

        for (; l <= n; l += l & -l) ft[l] += val;
    }

    int query(int l) {
        int ans = 0;
        for (; l > 0; l -= l & -l) ans += t[l];
        return ans;
    }

    int query(int l, int r) { return query(r) - query(l - 1); }
}

```

4.10 centroid decomposition

```

struct Centroid_Decomposition{
    vector<int> adj[N];
    int sz[N], vis[N], aux;
    int pa[N];

    int find(int u, int p, int Sz){
        int mx = 0, c = 0; sz[u] = 1;
        for (auto v : adj[u]){
            if (v == p || vis[v]) continue;
            c ^= find(v, u, Sz);
            sz[u] += sz[v];
            mx = max(mx, sz[v]);
        }
        if (max(mx, Sz - sz[u]) <= Sz/2)
            aux = p, c = u;
        return c;
    }

    void build(int u, int p, int Sz){
        aux = -1, u = find(u, -1, Sz);
        if (aux != -1) sz[aux] = Sz - sz[u];
        vis[u] = 1, sz[u] = Sz, pa[u] = p;
        for (auto v : adj[u])
            if (!vis[v]) build(v, u, sz[v]);
    }
} cd; // by NekoRolly

```

4.11 multi ordered set

```

// find_by_order(k) -> return iterator to the k-th element
// (0-indexed) - O(log n)
// order_of_key(num) -> # of items strictly smaller than num
// - O(log n)

// using less_equal cause that lower_bound works as
// upper_bound and viceversa
// also .find() does not work and you can only erase by
// iterator now
// using for example erase(upper_bound(x))
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;

```

```

template <typename T>
using ordered_set =
    tree<T, null_type, less_equal<T>, rb_tree_tag,
    tree_order_statistics_node_update>;

ordered_set<int> s;

```

4.12 hld

```

struct Graph {
    vector<vector<ll>> adj;
    vector<ll> padre, prof, heavy, head, pos, base, L, R;
    segmentTree st;
    ll curpos, n;
    ll dfs(ll u, ll p) {
        padre[u] = p;
        ll szu = 1;
        ll mxsz = 0;
        heavy[u] = 0;
        for (ll v : adj[u])
            if (v != p) {
                prof[v] = prof[u] + 1;
                ll szv = dfs(v, u);
                szu += szv;
                if (mxsz < szv) {
                    mxsz = szv;
                    heavy[u] = v;
                }
            }
        return szu;
    }

    void decompose(ll u, ll h, vector<ll> &a) {
        head[u] = h;
        L[u] = curpos;
        pos[u] = curpos++;
        base[pos[u]] = a[u - 1];
        if (heavy[u] != 0) decompose(heavy[u], h, a);
        for (ll v : adj[u])
            if (v != padre[u] && v != heavy[u]) decompose(v,
v, a);
        R[u] = curpos - 1;
    }

    void pupdate(ll u, ll val) {
        st.pupdate(pos[u], val);
    }

    void update(ll u, ll v, ll val) {
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            st.update(pos[head[v]], pos[v], val);
            v = padre[head[v]];
        }
        if (prof[u] > prof[v]) swap(u, v);
        st.update(pos[u], pos[v], val);
    }

    ll query(ll u, ll v) {

```

```

ll ans = INF;
while (head[u] != head[v]) {
    if (prof[head[u]] > prof[head[v]]) swap(u, v);
    ans = f(ans, st.query(pos[head[v]], pos[v]));
    v = padre[head[v]];
}
if (prof[u] > prof[v]) swap(u, v);
ans = f(ans, st.query(pos[u], pos[v]));
return ans;
}
ll subtreequery(ll u) {
    return st.query(L[u], R[u]);
}
void subtreeupdate(ll u, ll val) {
    st.update(L[u], R[u], val);
}
}
Graph(vector<ll> &a, vector<pll> &aristas, ll root =
1) : n(a.size()) {
    adj.assign(n + 1, vector<ll>()), padre.assign(n + 1,
0), prof.assign(n + 1, 0);
    heavy.assign(n + 1, 0), head.assign(n + 1, 0),
pos.assign(n + 1, 0);
    L.assign(n + 1, 0), R.assign(n + 1, 0),
base.resize(n);
    curpos = 0;
    for (auto [u, v] : aristas) {
        adj[u].pb(v);
        adj[v].pb(u);
    }
    prof[root] = 0;
    dfs(root, root);
    decompose(root, root, a);
    st = segmentTree(base);
}
};

```

4.13 segment tree

```

struct IterativeSegmentTree { // Index 0
    int n;
    ll ST[2 * MX];
    const ll NEUT = 0;
    void build(vector<ll> &a) {
        n = a.size();
        for (int i = n; i < (n << 1); i++) ST[i] = a[i - n];
        for (int i = n - 1; i > 0; i--) ST[i] = f(ST[i <<
1], ST[i << 1 | 1]);
    }
    ll f(ll x, ll y) { return max(x, y); }
    void update(int i, ll val) {
        for (ST[i += n] = val; i >= 1; i >>= 1) ST[i] = f(ST[i <<
1], ST[i << 1 | 1]);
    }
    ll query(int l, int r) { // [l, r)
        ll ansL = NEUT, ansR = NEUT;

```

```

        for (l += n, r += n; l < r; l >>= 1, r >>= 1) {
            if (l & 1) ansL = f(ansL, ST[l++]);
            if (r & 1) ansR = f(ST[--r], ansR);
        }
        return f(ansL, ansR);
    }
};

```

4.14 hld edge

```

struct Graph {
    vector<vector<pll>> adj;
    vector<ll> padre, wpadre, prof, heavy, head, pos, base,
L, R;
    segmentTree st;
    ll curpos;
    ll n;
    ll dfs(ll u, ll p, ll wp) {
        padre[u] = p;
        wpadre[u] = wp;
        ll szu = 1;
        ll mxsz = 0;
        heavy[u] = 0;
        for (auto [v, w] : adj[u])
            if (v != p) {
                prof[v] = prof[u] + 1;
                ll szv = dfs(v, u, w);
                szu += szv;
                if (mxsz < szv) mxsz = szv, heavy[u] = v;
            }
        return szu;
    }
    void decompose(ll u, ll h) {
        head[u] = h;
        L[u] = curpos;
        pos[u] = curpos++;
        base[pos[u]] = wpadre[u];
        if (heavy[u] != 0) decompose(heavy[u], h);
        for (auto [v, w] : adj[u])
            if (v != padre[u] && v != heavy[u]) decompose(v,
v);
        R[u] = curpos - 1;
    }
    void pupdate(ll u, ll v, ll val) {
        if (u == padre[v]) swap(u, v);
        st.pupdate(pos[u], val);
    }
    void update(ll u, ll v, ll val) {
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            st.update(pos[head[v]], pos[v], val);
            v = padre[head[v]];
        }
        if (prof[u] > prof[v]) swap(u, v);
        if (u != v) st.update(pos[u] + 1, pos[v], val);
    }
};

```

```

}
ll query(ll u, ll v) {
    ll ans = INF;
    while (head[u] != head[v]) {
        if (prof[head[u]] > prof[head[v]]) swap(u, v);
        ans = f(ans, st.query(pos[head[v]], pos[v]));
        v = padre[head[v]];
    }
    if (prof[u] > prof[v]) swap(u, v);
    if (u != v) ans = f(ans, st.query(pos[u] + 1,
pos[v]));
    return ans;
}
ll subtreequery(ll u) {
    if (L[u] == R[u]) return 0;
    return st.query(L[u] + 1, R[u]);
}
void subtreeupdate(ll u, ll val) {
    if (L[u] == R[u]) return;
    st.update(L[u] + 1, R[u], val);
}
}
Graph(vector<pair<pll, ll>> &aristas, ll root = 1) :
n(aristas.size() + 1) {
    adj.assign(n + 1, vector<pll>());
    padre.assign(n + 1, 0), wpadre.assign(n + 1, 0),
prof.assign(n + 1, 0);
    heavy.assign(n + 1, 0), head.assign(n + 1, 0),
pos.assign(n + 1, 0);
    L.assign(n + 1, 0), R.assign(n + 1, 0),
base.resize(n);
    curpos = 0;
    for (auto [edge, w] : aristas) {
        auto [u, v] = edge;
        adj[u].pb({v, w});
        adj[v].pb({u, w});
    }
    prof[root] = 0;
    dfs(root, root, 0LL);
    decompose(root, root);
    st = segmentTree(base);
}
};

```